

# Go Language Fundamentals

*AI Tutor - Golang - Lesson 1*

## What is Go?

Go (or Golang) is an open-source programming language created at Google, designed for simplicity, fast compilation, and strong support for concurrent programs. It is widely used for backend servers, APIs, and cloud infrastructure.

## A Simple Go Program

Every Go program starts with a package declaration and a main function:

- `package main`
- `import "fmt"`
- `func main() { fmt.Println("Hello, Go!") }`

## Why Go for Backends

- Compiles to a single fast, native binary.
- Built-in support for concurrency (goroutines).
- Simple, readable syntax with strong typing.
- Popular frameworks like Gin make building REST APIs fast.